Chapter2 Simple Machines

Objectives You will be able to:

- Explain how machines help us do work.
- Define efficiency, and state how it is calculated.
- Define mechanical advantage, and state how it is calculated.
- Compare the three classes of levers.
- Explain how an inclined plane changes force.
- Relationship between number of turns and teeth of gears.
- Describe two ways that a wheel and axle can be used.

Machine

When you hear the word machine, do you think of power tools or construction equipment, like the ones pictured below? While both of these examples are machines, you might be surprised to learn that devices as simple as hammers and screws are also machines.



A **machine** is any device that makes work easier by changing a force. Work is done whenever a force moves an object over a distance. The amount of work done is represented by the equation: Work = Force x Distance

When you use a machine, you apply force to the machine. This force is called the input force. The machine, in turn, applies force to an object. This force is called the output force. The output force may or may not be the same as the input force.

The force you apply to the machine is applied over a given distance, called the input distance. The force applied by the machine to the object is also applied over a distance, called the output distance. The output distance may or may not be the same as the input distance.

A compound machine is a machine that consists of more than one simple machine.

How do Machines make Work easier?

Contrary to popular belief, machines do not increase the amount of work that is done. They just change how the work is done. Machines make work easier by increasing the amount of force that is applied, increasing the distance over which the force is applied, or changing the direction in which the force is applied.

Increasing Force

Examples of machines that increase force are steering wheels and pliers

Increasing Distance

Examples of machines that increase the distance over which force is applied are leaf rakes and hammers.

Changing the Direction of Force

Some machines change the direction of the force applied by the user. They may or may not also change the strength of the force or the distance over which the force is applied. Two examples of machines that work this way are the claw ends of hammers and flag pulleys.

Simple machine includes levers, inclined plane, gears, pulley, wheel and axle, and wedge.



Lesson 1 Lever

Objective: You will be able to:

-Define lever
-Describe principle of conservation of work
-Use lever in real life.
-Calculate mechanical advantage.

1-Lever

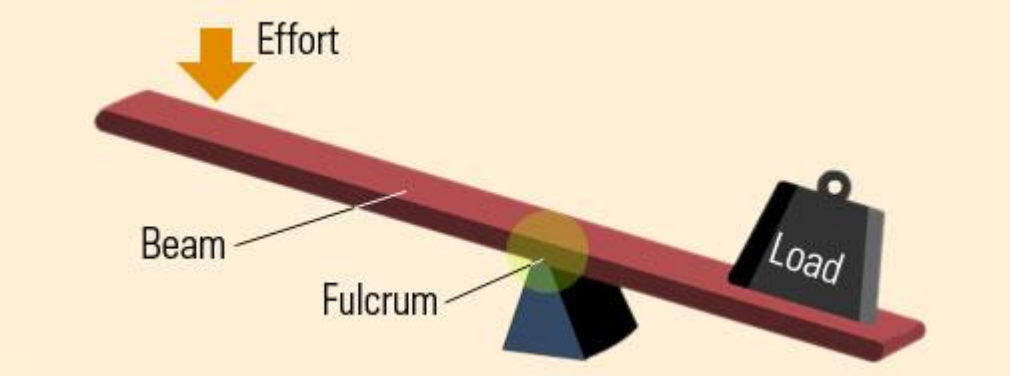
A **lever** is a rigid rod, pivoted about a fixed point or axis, which is known as a fulcrum. A lever amplifies an input force to provide a greater output force.

In this diagram, a boy uses a see-saw to lift up another boy.



Parts of a lever

A typical lever consists of a solid board or rod that can pivot about a point or fulcrum. A force or effort is applied, resulting in moving or applying force to a load. The distance from the applied force or effort force to the fulcrum is called the effort arm and the distance from the load to the fulcrum is called the load arm.



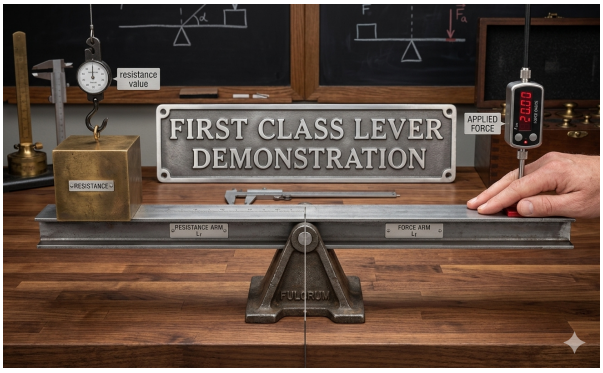
- ***Fulcrum*** or pivot – the point about which the lever rotates.
- ***Load*** – the force applied by the lever system.
- ***Effort*** – the force applied by the user of the lever system.

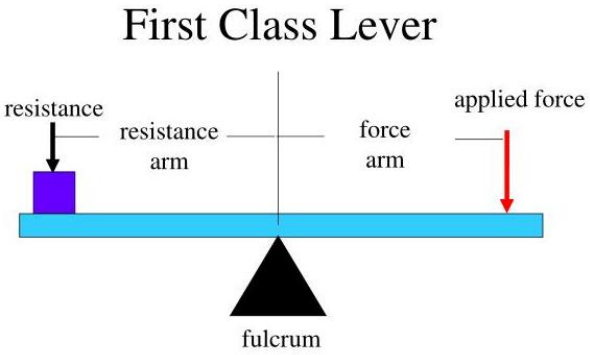
2- Classes of levers

Levers are classified by the relative positions of the fulcrum and the input and output forces. It is common to call the input force or *the effort* or the *output force*, *the load* or *the resistance.*

First Class Levers

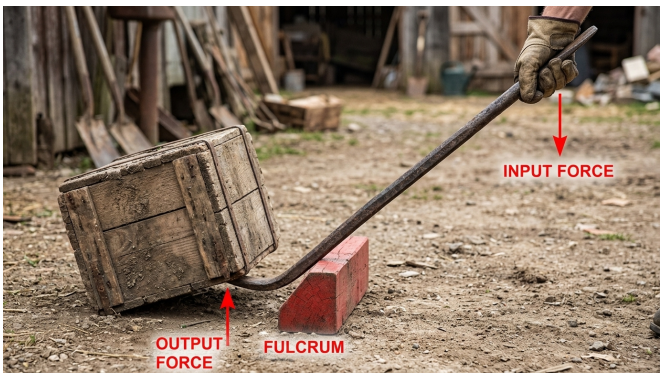
This is the most common type of lever, with the fulcrum is between the effort on one side and the load on the other.

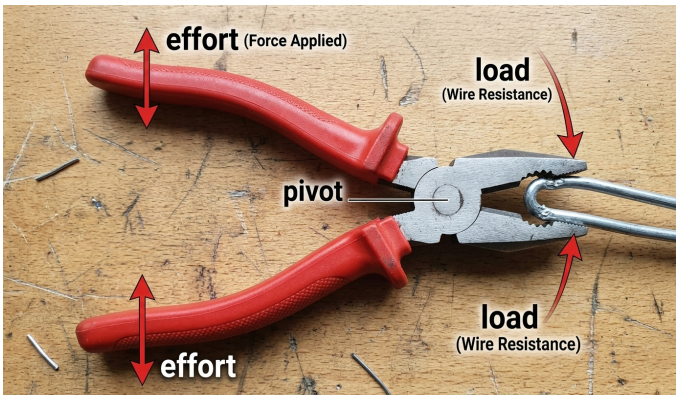


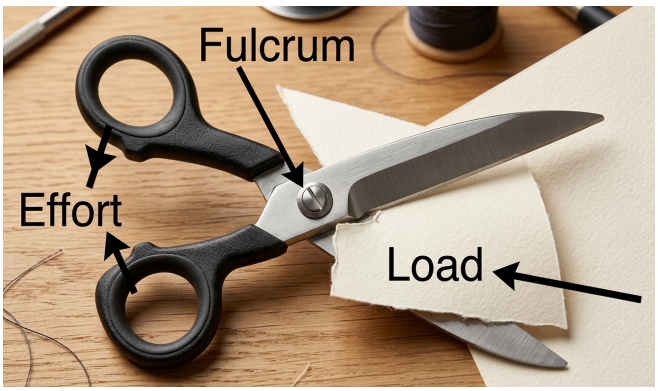


Examples of common tools (and other items) that use a type 1 lever include: see-saw, scissor, pliers





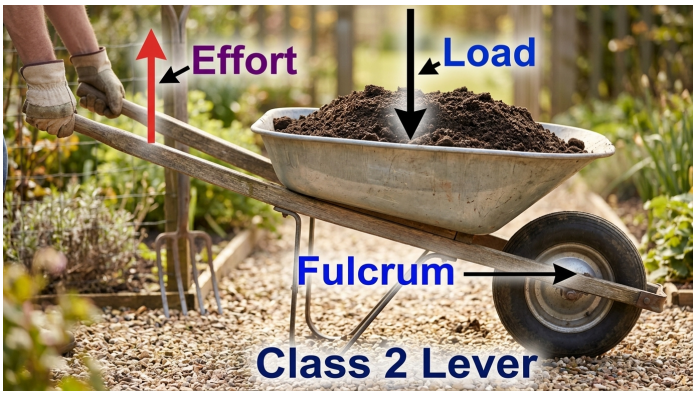




Second Class Levers

- With a Class Two Lever, the fulcrum is at one end, the effort is at the other end and the load is between.

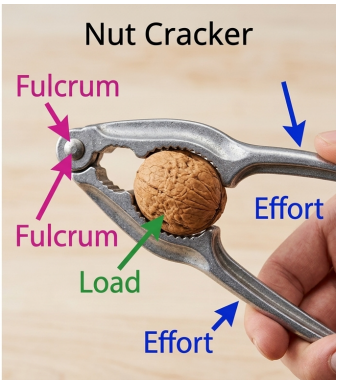
- A Class Two Lever allows a large load to be than the lifted by a smaller effort. Because the load is always closer to the fulcrum, the effort is always less load



Examples of common tools that use a second class lever include: Wheel Barrow, Bottle opener, nut cracker,

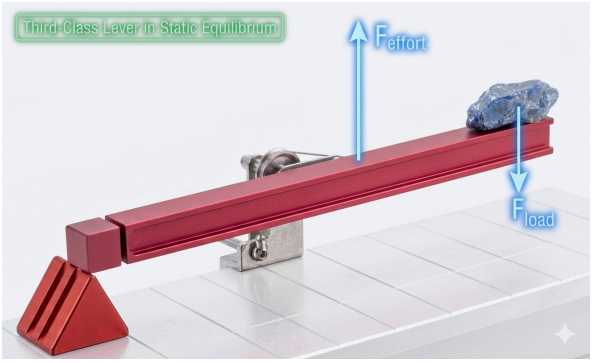




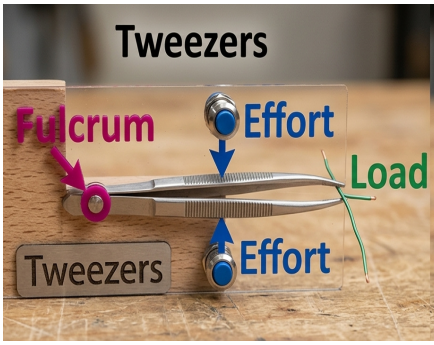


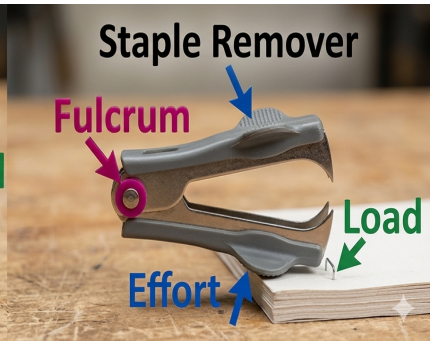
Third Class Levers

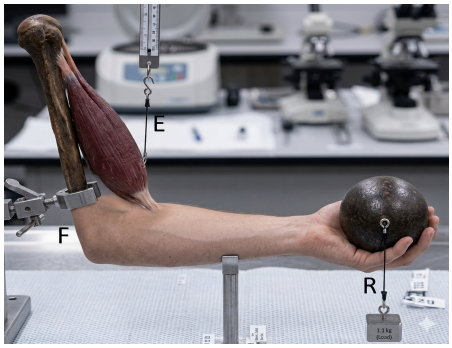
- With a Class Three Lever, the pivot is at one end, the load is at the other and the effort is in the middle.
- A Class Three Lever allows a small load to be lifted by a larger effort



Example of third class lever: Fishing rod, tong, tweezers, stapler.







3. Mechanical Advantage

Mechanical advantage: is the number of times a machine multiplies the effort (input force).

The general formula for the mechanical advantage (MA) of levers:

$$MA = \frac{Load}{Effort} = \frac{F_R}{F_E} = \frac{d_E}{d_R}$$

d_E is the effort arm length from fulcrum. d_R is the arm length of load from fulcrum

-The mechanical advantage of a machine may be greater than, less than, or equal to 1, depending on the type of machine.

- MA is a ratio, so it has no unit. Load and effort are forces. Load is the force that the machine does. Effort is the force that applies to the machine.

Comparing Classes of Levers

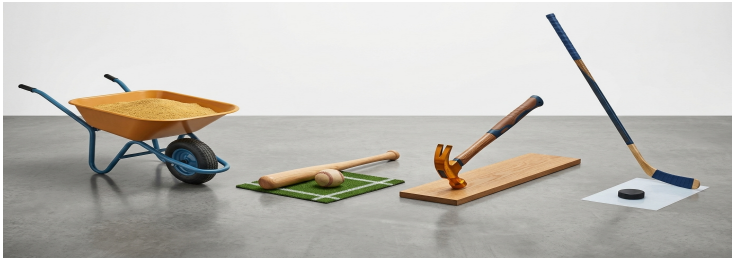
All three classes of levers make work easier, but they do so in different ways.

- When the input and output forces are on opposite sides of the fulcrum, the lever changes the direction of the applied force. This occurs only with a first-class lever.
- When both the input and output forces are on the same side of the fulcrum, the direction of the applied force does not change. This occurs with both second- and third-class levers.
- When the input force is applied farther from the fulcrum, the input distance is greater than the output distance, so the ideal mechanical advantage is greater than 1. This always occurs with second-class levers and may occur with first-class levers.
- When the input force is applied closer to the fulcrum, the input distance is less than the output distance, so the ideal mechanical advantage is less than 1. This always occurs with third-class levers and may occur with first-class levers.
- When both forces are the same distance from the fulcrum, the input distance equals the output distance, so the ideal mechanical advantage equals 1. This occurs only with first class-lever.

Practice

1- Tell if the following simple machines are 1st, 2nd, or 3rd class levers.



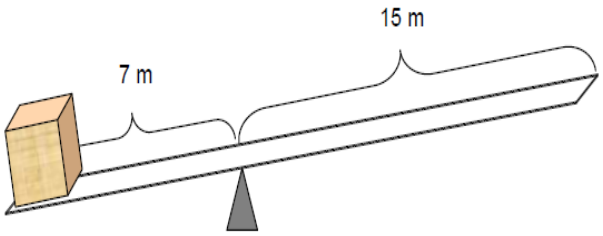


2. The mechanical advantage of a lever is 6. How much **force** needs to be exerted on the lever to lift an object with a gravitational force of 320 N?

3- A broom with an input arm length of 0.4 meters has a mechanical advantage of 0.5. What is the length of the output arm?

4. A child's toy rake is held so that its output arm is 0.75 meters. If the mechanical advantage is 0.33, what is the input arm length?

5- Calculate the mechanical advantage of this lever.



6-Using a lever, a person applies 60N of force and moves the lever 1 meter.
This moves a 200 N rock at the other end by 0.2 meters. What is the efficiency of the machines?